Warm-Up Quiz — Day 12

Quadratic Forms, the χ^2 Distribution, and Linear Restrictions

Complete before lecture. 10 minutes.

Name: ____________________

Today we study hypothesis testing: the Wald test, the F-test, and how to impose linear restrictions $\boldsymbol{R}'\boldsymbol{\beta} = \boldsymbol{r}$. This quiz reviews the algebra of quadratic forms and the chi-squared distribution that underpin these tests.

1. **Quadratic Forms.**
   A quadratic form is an expression $\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x}$ where $\boldsymbol{A}$ is a symmetric matrix.

   (a) Let $\boldsymbol{x} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\boldsymbol{A} = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$. Compute $\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x}$.

   (b) If $\boldsymbol{z} \sim N(\boldsymbol{0}, \boldsymbol{I}_k)$, then $\boldsymbol{z}'\boldsymbol{z} \sim \chi^2_k$. What distribution does $\boldsymbol{z}'\boldsymbol{z}$ follow when $k = 3$?

2. **Standardizing a Normal Vector.**
   If $\boldsymbol{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, we can standardize: $\boldsymbol{z} = \boldsymbol{\Sigma}^{-1/2}(\boldsymbol{X} - \boldsymbol{\mu}) \sim N(\boldsymbol{0}, \boldsymbol{I})$.
   Then $(\boldsymbol{X} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\boldsymbol{X} - \boldsymbol{\mu}) = \boldsymbol{z}'\boldsymbol{z} \sim \chi^2_k$.

   (a) If $\hat{\beta} \sim N(\beta, \sigma^2/n)$ (scalar), write $(\hat{\beta} - \beta)'(\sigma^2/n)^{-1}(\hat{\beta} - \beta)$ in simplified form.

   (b) What distribution does this quantity follow?

3. **Writing Hypotheses as Linear Restrictions.**
   Write each hypothesis in the form $\boldsymbol{R}'\boldsymbol{\beta} = \boldsymbol{r}$, where $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3)'$.

   (a) $H_0 : \beta_2 = 0$ (single exclusion).

   (b) $H_0 : \beta_1 = \beta_2$ (equality of coefficients).

(c) $H_0 : \beta_1 = 0$ and $\beta_2 = 0$ (joint exclusion).

4. The Wald Statistic (Preview).
The Wald test for $H_0 : \boldsymbol{R}'\boldsymbol{\beta} = \boldsymbol{r}$ is:

$$W = (\boldsymbol{R}'\hat{\boldsymbol{\beta}} - \boldsymbol{r})' \left[\boldsymbol{R}'\hat{\boldsymbol{V}}\boldsymbol{R}\right]^{-1} (\boldsymbol{R}'\hat{\boldsymbol{\beta}} - \boldsymbol{r})$$

(a) For the scalar case $H_0 : \beta_1 = 0$, show that W reduces to $(\hat{\beta}_1/\mathrm{se}(\hat{\beta}_1))^2$, i.e., the square of the t-statistic.

Answer Key — Day 12

1. (a) $\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x} = (2,1)\begin{pmatrix}2\\4\end{pmatrix} = 4+4 = 8.$

(b) χ^2_3.

2. (a) $\frac{n}{\sigma^2}(\hat{\beta}-\beta)^2 = \left(\frac{\hat{\beta}-\beta}{\sigma/\sqrt{n}}\right)^2.$

(b) χ^2_1 (square of a standard normal is chi-squared with 1 degree of freedom).

3. (a) $\boldsymbol{R}' = (0,0,1,0)$, $\boldsymbol{r} = 0$.

(b) $\boldsymbol{R}' = (0,1,-1,0)$, $\boldsymbol{r} = 0$.

(c) $\boldsymbol{R}' = \begin{pmatrix}0 & 1 & 0 & 0\\0 & 0 & 1 & 0\end{pmatrix}$, $\boldsymbol{r} = \begin{pmatrix}0\\0\end{pmatrix}$.

4. (a) Here $\boldsymbol{R}' = (0,1,0,0)$ picks out $\hat{\beta}_1$, $\boldsymbol{r} = 0$, and $\boldsymbol{R}'\hat{\boldsymbol{V}}\boldsymbol{R} = \hat{V}_{11} = [\text{se}(\hat{\beta}_1)]^2$. So $W = \hat{\beta}_1^2/[\text{se}(\hat{\beta}_1)]^2 = t^2$.